Astronomical Bycatch: progress report

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1 Summary

The aim is to get light curves for the stars in archival images using the new extraction API in our private EXOTIC fork (`exotic.extract`). This report covers the first day of that work:

- Code to search and download raw data from the ESO Science Archive, and to find archive sequences that cover a known transit.
- A pipeline that plate-solves frames locally against Gaia DR3, runs `extract_sequence` on every usable star in the field, fits the target's transit and writes all light curves to JSON.
- **Validation:** on the HAT-P-32 b MicroObservatory night, the fitted mid-transit times agree with EXOTIC's own result to within 1–3 min (0.7–2.1σ).
- **Wide-field recovery:** POTS-1 b recovered in archival ESO/MPG 2.2 m WFI data, among 801 field stars on one detector: T_{mid} within 4.7 ± 4.0 min of the ephemeris and $R_p/R_s = 0.166 \pm 0.010$ (catalogue 0.164).
- Two problems found in EXOTIC on the way: its sky-background estimate fails on calibrated (floating-point) frames, and its quick transit fit can stall at absolute BJD times (Section 5).

2 Data sources

ESO Science Archive (raw frames). Metadata was queried through ESO's catalogue query service, and transiting planets through the NASA Exoplanet Archive. The two optical wide-field imagers, WFI and OmegaCAM, have 182 long single-pointing sequences between them (at least 40 frames over at least 2 hours). Only one catches a known transit: **POTS-1 b, WFI, night of 2007-05-10**. It has 310 Rc-band frames of 26 s over 8.2 h, covering the full transit with ∼2.5 h of data before and 2.8 h after. The night has 24 biases, 4 twilight flats and 10 screen flats; the night is 46 GB decompressed. POTS-1 is faint ($G = 17.3$) but its transit is deep, and the field is crowded with background stars.

MicroObservatory frames. David Lawley's repository of raw MicroObservatory frames covering seven hot-Jupiter targets. The HAT-P-32 night is byte-identical to EXOTIC's own sample data, which makes it a direct check against EXOTIC.

3 Code and storage

`access/` Searches the ESO archive, matches sequences to known transits, and downloads a target's frames and calibrations (`python -m access.fetch targets/<t>.json`). Downloads can be restarted. Each target is a small JSON file in `targets/`.

`pipeline/astrometry.py` Local plate solving against Gaia DR3, so no image is uploaded to an external service. Raw WFI and MicroObservatory frames carry no sky coordinates, and the MicroObservatory header pointing can be ∼0.3° off.

`pipeline/microobs.py`, `pipeline/wfi.py` Pipelines for each instrument. WFI handles 8 detectors per frame: overscan trimming, master bias and flat, and a separate plate solve per detector.

`pipeline/store.py` JSON results, one file per sequence (and per detector). Each holds the observation details, per-frame times and airmass, and a light curve for every star keyed by Gaia DR3 ID. The layout maps onto the existing kdb+ schema (`sources`, `obs_sessions`, `observations`).

Data and results stay on the department compute server; nothing large is kept on laptops. Code is on the `huw-leggate/claude-md` branch of `creanero/bycatch`.

4 Results

4.1 HAT-P-32 b: validation

142 frames of 60 s. The plate solve matched 184 Gaia stars to 1.5 arcsec, and 83 stars were photometered.

Method. Each star's flux is measured in a circular *photometric aperture* centred on the star, with the sky level taken from a surrounding annulus. The aperture radius is set as a multiple of the stellar FWHM measured on the first frame. The target's relative light curve is its flux divided by the summed flux of the comparison stars. That curve is fitted with EXOTIC's least-squares `fit_transit`, after dividing out a linear trend with airmass. Times are measured from the start of the night rather than as absolute BJD, the fit starts from a coarse grid search, and points more than 4σ from the model are clipped (Section 5). The fit has three free parameters:

- the mid-transit time T_{mid};
- the radius ratio R_p/R_s, the planet's physical radius divided by its host star's radius, which sets the transit depth (depth $\approx (R_p/R_s)^2$);
- a flux normalisation.

The orbit (period, scaled semi-major axis a/R_s, inclination) and the stellar limb darkening are held fixed. R_p/R_s has nothing to do with the photometric aperture radius. The aperture is only a measurement setting, and a well-behaved reduction should return the same R_p/R_s whatever aperture is used.

What is compared. We reduced the same night three times, with photometric apertures of 1.0, 1.4 and 1.9× the FWHM, and fitted each light curve separately (Table 1). Each fit is compared against two independent references:

1. **Timing.** EXOTIC's own full reduction of these same frames gives $T_{\mathrm{mid}} = 2458107.71380\,\mathrm{BJD_{TDB}}$. This is the value EXOTIC's validation tests check against. Matching it shows our extraction and timing reproduce EXOTIC's.
2. **Depth.** The literature value $R_p/R_s = 0.149$, from the HAT-P-32 b parameters in EXOTIC's configuration for this dataset (taken from the NASA Exoplanet Archive). EXOTIC's validation tests don't check R_p/R_s: in its fit R_p/R_s is degenerate with the orbital inclination, so it isn't stable enough to test against. So depth is compared with the published value instead.

Outcome.

- **Timing:** the three fitted mid-transit times are 1.1 to 3.1 min earlier than EXOTIC's, with uncertainties of about 1.5 min (0.7–2.1σ). Some offset is expected: EXOTIC uses its own nested-sampling fit, different limb darkening and a different detrending.

Table 1: HAT-P-32 b transit fits for three photometric aperture sizes. Each row is a separate reduction of the same 142 frames. *Aperture radius* is the photometric aperture in pixels and in multiples of the stellar FWHM. *Scatter* is the point-to-point scatter of the relative light curve. T_{mid} *offset* is the fitted mid-transit time minus EXOTIC's result for this night (2458107.71380 $\mathrm{BJD_{TDB}}$). *Fitted* R_p/R_s is the planet-to-star radius ratio from the fit; the literature value is 0.149.

Photometric aperture radius (×FWHM)	(px)	Scatter	T_{mid} offset (min)	Fitted R_p/R_s (planet / star)
1.0	2.5	0.53%	-3.1 ± 1.5	0.151 ± 0.003
1.4	3.5	0.48%	-1.9 ± 1.4	0.158 ± 0.003
1.9	4.7	0.57%	-1.1 ± 1.6	0.157 ± 0.003

- **Depth:** the fitted planet-to-star radius ratio is 0.151 to 0.158, 1–6% above the literature 0.149. The most likely cause is the limb darkening, held at generic approximate coefficients rather than ones computed for HAT-P-32 in this filter. That changes the transit's shape, and with it the depth the fit needs. It has little effect on T_{mid}, which depends on the transit's symmetry.

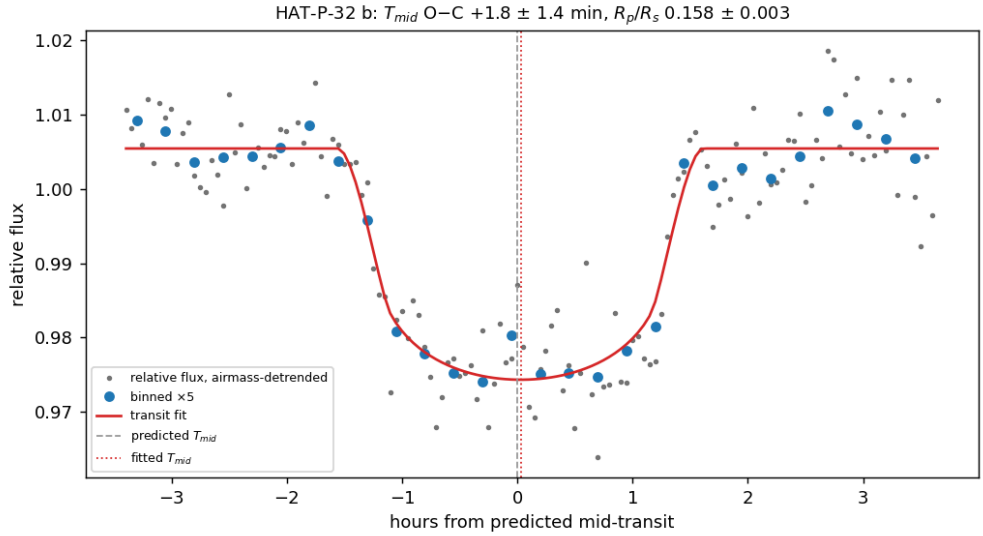


Figure 1: HAT-P-32 b relative light curve (grey points) with the fitted transit model (red), from the reduction with a photometric aperture radius of 1.4×FWHM (3.5 px). The x-axis is hours from the mid-transit time predicted by the published ephemeris (grey dashed line); the red dotted line is the fitted T_{mid}, 1.75 min after that prediction and 1.9 min before EXOTIC's result for this night. The fitted R_p/R_s is 0.158 (Table 1).

4.2 WASP-36 b

55 frames of 60 s. Plate solving and frame alignment worked, but the target ($G = 12.6$) is limited by sky noise: the best scatter is 2.3% per point, with a photometric aperture radius of 1.0×FWHM (2.3 px). Scatter fell steadily with smaller apertures, as expected for a faint star dominated by sky noise (6.2% at the default 1.9×FWHM). The transit is ∼1.8% deep. The fit (Figure 2) gives T_{mid} $+15.4 \pm 6.3$ min from the ephemeris prediction and $R_p/R_s = 0.152 \pm 0.022$ (literature 0.134). Treat this as provisional. A 66-minute gap leaves only 3 frames before ingress, so the out-of-transit level is set almost entirely after egress, and a timing offset of 2.4σ may well be systematic. With the default 1.9×FWHM aperture the fit gives -30 ± 27 min and 0.17 ± 0.07.

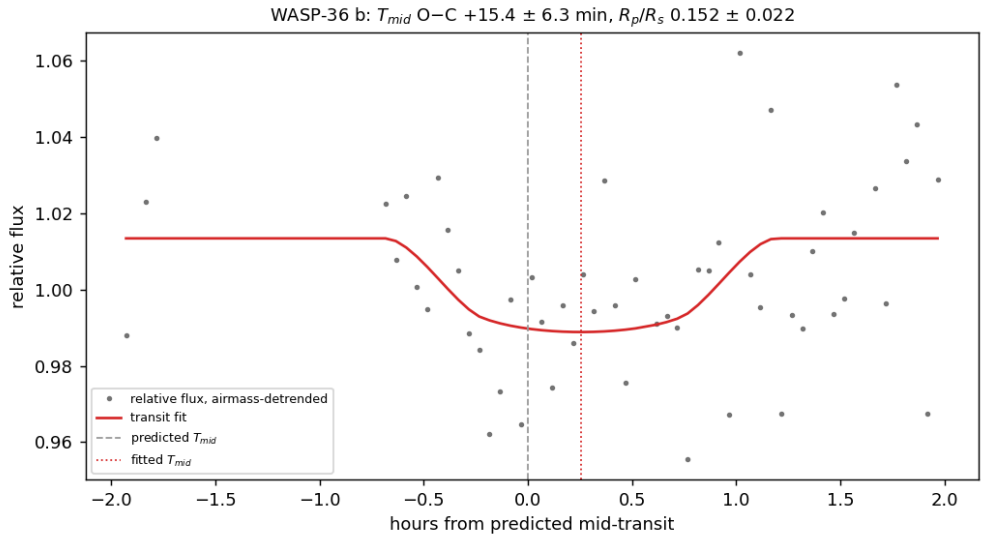


Figure 2: WASP-36 b relative light curve (grey points) with the fitted transit model (red), from the reduction with a photometric aperture radius of 1.0×FWHM (2.3 px). The dashed grey line is the predicted mid-transit time and the dotted red line the fitted one. Only three points precede the transit.

4.3 POTS-1 b, WFI

- **Calibration:** gain and read noise were measured for each detector from bias and flat pairs, because the headers record gain for only two detectors and read noise for none. Gain came out at 1.94–2.28 e^-/ADU and read noise at 5.6–6.2 e^-, consistent with WFI's documented values.
- **Astrometry:** all 8 detectors plate-solved, with 36–163 Gaia stars each and residuals of 0.06–0.11 arcsec. POTS-1 is on chip 3.

First extraction (failed). 801 stars down to $G = 18.5$ were photometered across all 310 frames, with a photometric aperture radius of 1.9×FWHM (10.1 px). The result is shown for information in Figure 3. The target's relative light curve scatters by 27% from point to point. That's far above the ∼1% photon noise expected for a $G = 17.2$ star in a 26 s exposure on a 2.2 m telescope, and it swamps the ∼2.5% transit. Every star on the chip is affected: scatter is 1.8% at $G = 12$, 4.4% at $G = 15$ and over 20% at $G \geq 17$, against predicted uncertainties of 0.06–0.9%. Star tracking was fine: the target stays within 0.4 px of the comparison stars.

There were two causes:

- **Sky background (EXOTIC).** `skybg_phot` takes the sky level as the statistical *mode* of the pixel values in a box around each star. The box includes the star itself, not just the annulus. On raw integer frames (MicroObservatory) the mode works, which is why HAT-P-32 validates. On bias- and flat-corrected frames, which are floating point, almost every value is unique, so the "mode" is an arbitrary pixel. For the target it jumped between 28 and 62 ADU from frame to frame, against a true sky of ∼50 ADU. Over the 320-pixel aperture that is a 20–30% flux error. Workaround: round calibrated frames to whole ADU before extraction. That adds 0.3 ADU of rounding noise against 7 ADU of sky noise.
- **Fixed orbit (ours).** The NASA Exoplanet Archive has no tabulated a/R_s for POTS-1 b, and the code defaulted it to zero, so the transit model was degenerate (hence the infinite uncertainty in Figure 3). It is now derived from the semi-major axis and the stellar radius.

Corrected extraction. Re-run with both fixes at two photometric aperture radii. Each run takes ∼30 min on 12 cores. Ephemeris and catalogue values are from the NASA Exoplanet Archive.

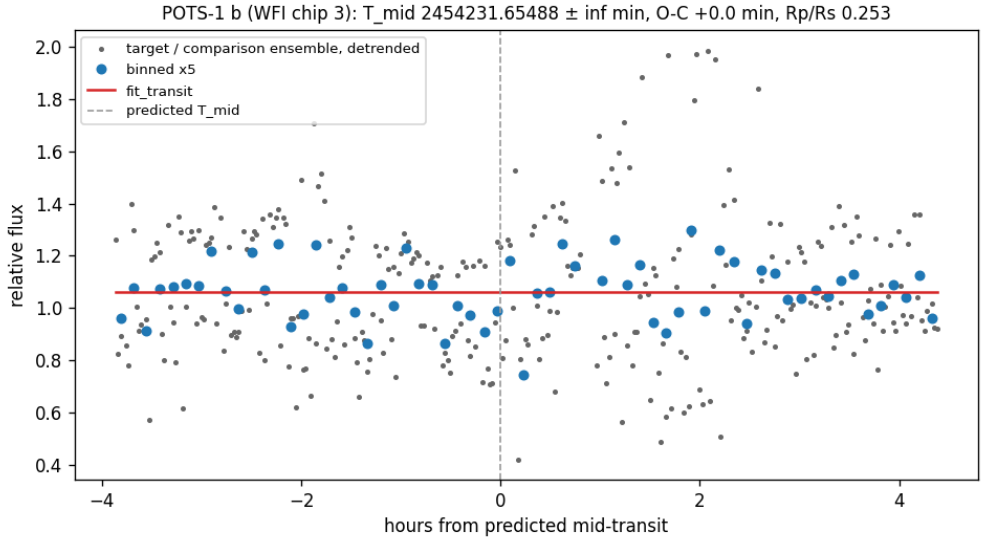


Figure 3: First, failed extraction of POTS-1 b (WFI chip 3), shown for information. Grey points are the target's relative flux per frame, blue points are 5-frame bins, and the red line is the failed fit. The 27% point-to-point scatter comes from the unstable sky-background estimate on calibrated (floating-point) frames. The fit is meaningless because a/R_s was zero. Both problems are fixed in the re-run.

Note that these WFI data are the survey in which POTS-1 b was discovered, so this tests whether we recover a known result, not an independent detection.

Table 2: POTS-1 b transit fits from WFI chip 3 (310 frames, 801 stars photometered) for two photometric aperture radii. T_{mid} *offset* is relative to the ephemeris prediction (uncertainty 0.6 min). *Fitted* R_p/R_s is the planet-to-star radius ratio; the catalogue value is 0.164. *Scatter* is the rms of the fit residuals. *Clipped* is the number of frames dropped as $> 4\sigma$ outliers.

Photometric aperture radius (×FWHM)	(px)	Scatter	T_{mid} offset (min)	Fitted R_p/R_s (planet / star)	Clipped
1.2	6.4	2.3%	$+4.7 \pm 4.0$	0.166 ± 0.010	2
1.9	10.1	4.1%	$+5.8 \pm 7.6$	0.159 ± 0.018	3

Both reductions recover the transit at the predicted time (within 1.2σ) and with the catalogue depth (Table 2, Figure 4). The smaller aperture halves the scatter, as for the faint MicroObservatory target. Frames between about +1 and +3 h after mid-transit are noisier than the rest; the cause (transparency, or a neighbouring star) hasn't been checked yet.

5 Issues and next steps

- **Speed:** centroid fitting in `extract_frame` takes ∼70 ms per star per frame, which dominates run time (about 55 s per WFI frame for 800 stars). Running in parallel works around it for now. This is a useful benchmark for the GPU work in EXOTIC.
- **EXOTIC sky background:** `skybg_phot`'s modal estimate fails on calibrated, floating-point frames (Section 4.3). We work around it by rounding, but the proper fix belongs in EXOTIC: estimate the sky from the annulus pixels only, with a sigma-clipped median, so it works whatever the data type. EXOTIC's own pipeline is affected whenever it applies darks or flats.
- **EXOTIC quick fit at absolute BJD:** `quick_fit.fit_transit` passes BJD ($\sim 2.45 \times 10^6$) straight to `curve_fit`, whose finite-difference step scales with the parameter value. The step on T_{mid} is then about 50 min, wider than the ingress. On the noisy POTS-1 data the fit stalled

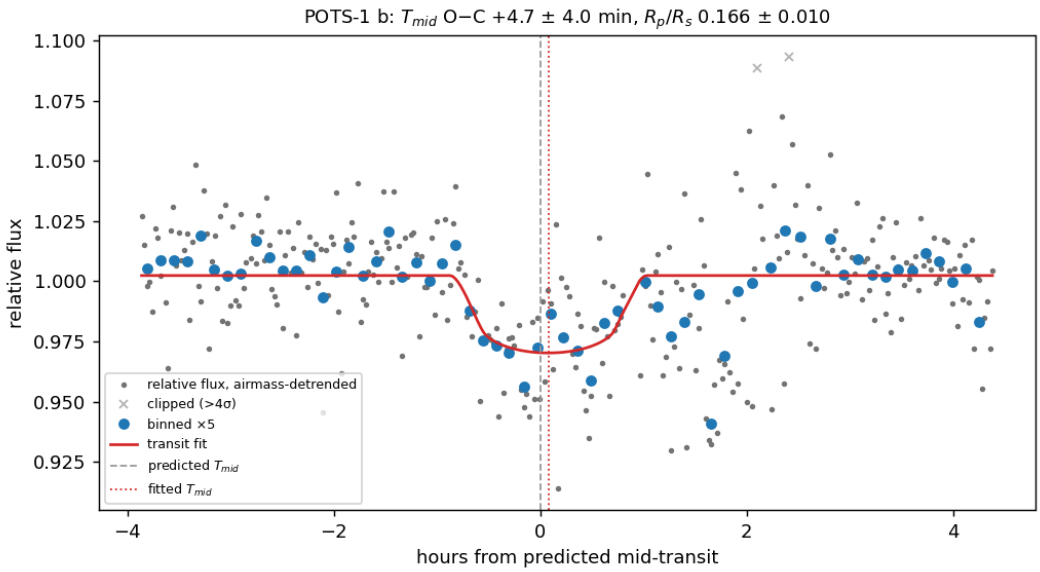


Figure 4: POTS-1 b relative light curve from WFI chip 3 with a photometric aperture radius of 1.2×FWHM (6.4 px). Grey points are single 26 s frames, blue points 5-frame bins, crosses frames clipped as $> 4\sigma$ outliers. The red line is the transit fit. The grey dashed and red dotted lines mark the predicted and fitted mid-transit times.

at its starting values. On HAT-P-32 and WASP-36 it converged only partly and overstated its uncertainties. Earlier numbers in this report have been refitted. Fix used here: fit times relative to the start of the night, from a grid-search start. It would be simple to do the same inside `fit_transit`.

- **Photometric aperture size:** the default FWHM-scaled aperture is too large for faint, sky-limited stars. It should be tuned per star, or chosen by sweeping, for bycatch at scale.
- **Limb darkening:** fixed approximate coefficients bias the fitted R_p/R_s. Timing is unaffected.
- **Next:** check the noisy interval in the POTS-1 light curve; run the remaining MicroObservatory targets; look for variability among the field stars in the JSON results; then scale the search to OmegaCAM and the other WFI sequences.